

BENJAMIN MASTERS

Graduate Researcher, Hearing Lab
Department of Systems Design Engineering
University of Waterloo

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EDUCATION

University of Waterloo	Waterloo, Ontario
Ph.D., Biomedical Engineering	Expected 2028
M.A.Sc., Systems Design Engineering	2024
Graduate Certificate in University Teaching	2026
Purdue University	West Lafayette, Indiana
B.S.E., Acoustical Engineering	2022

FELLOWSHIPS

Canada Graduate Research Scholarship – Doctoral	2026 – 2028
Project: Eye-tracking based assessments of effort in conversational interactions	C\$120,000
Awarded by the Natural Sciences and Engineering Research Council of Canada	

RESEARCH APPOINTMENTS

University of Waterloo	Waterloo, Ontario
Graduate Researcher, Hearing Lab	2022 – Present
- Studying the ways people adapt to difficulties in conversation to develop approaches for measuring cognitive effort during communication. - The long-term goal of this work is to identify ways to assess the effectiveness of hearing aids in realistic settings. - Developed expertise in pupillometry, conversational behavior, speech production, hearing science, acoustic and bio-signal processing, systems design, and precision calibration and synchronization of multi-modal signals.	
Purdue University	West Lafayette, Indiana
Undergraduate Researcher, Experimental Amplification Research Lab	2021 – 2022
- Assessed and improved the acoustical performance of an open-source hearing aid platform.	

INDUSTRY EXPERIENCE

Bose Corporation	Framingham, Massachusetts
Acoustic Engineer Intern	2022
- Developed an automated algorithm and testing procedure to acoustically detect air leaks from loudspeaker enclosures.	
Starkey Hearing Technologies	Eden Prairie, Minnesota
Acoustic Engineer Intern	2021
- Adapted vibration testing procedures for electroacoustic transducers to improve coherence of optical measurements.	

TEACHING APPOINTMENTS

University of Waterloo	Waterloo, Ontario
Course Instructor	Spring 2025 & 2026
• Served as the sole instructor for a core undergraduate engineering course, Linear Systems & Signals, with ~80 students. • Restructured course content such that it could be delivered with a cohesive narrative throughout the term. • Designed and delivered all lecture and assessment materials and managed two teaching assistants. • Achieved an average teaching evaluation of 4.7/5 across content and delivery.	
Teaching Assistant	2022 – Present
- Acted as a teaching assistant for several courses over 10+ academic semesters. Courses TA'd: Introduction to Biomedical Design, Biomedical Signal Processing, Linear Systems & Signals (x5), Computational Neuroscience, Simulating Neurobiological Systems, Communication in Biomedical Engineering - Performed duties such as providing design guidance, marking assignments and exams, holding office hours, and developing and hosting weekly interactive tutorials, labs, and guest lectures.	

- Graded assignments and exams for a course in Acoustics and Noise Control.

SERVICE

Acoustical Society of America

- Student Council Representative - Technical Committee on Signal Processing 2025 – 2027
- Session Co-chair - Undergraduate and Graduate Programs in Acoustics 2026

Internal/Institutional

- Organizing Committee - Symposium on Aging Research 2025 – 2026
- Biomedical Engineering Program Representative - Various Graduate Studies Fairs 2025 – 2026
- Committee Member - Systems Design/Biomedical Engineering Teaching Assistant Spaces 2025
- Treasurer - Purdue Student Chapters of the Acoustical Society of America and Audio Engineering Society 2020 – 2022
- Vice-President and Treasurer - Purdue University Multidisciplinary Engineering Student Society 2019 – 2021

AWARDS

- International Symposium on Auditory and Audiological Research Scholarship 2025
- Biomedical Engineering Research Day Best Presentation Award 2025
- Research Institute for Aging Best Presentation Award 2025
- 1st Place - International Student Challenge in Signal Processing - Acoustical Society of America 2024
- Canadian Acoustical Association Student Travel Award 2024
- 1st Place - International Student Challenge in Signal Processing - Acoustical Society of America 2023

SCHOLARSHIP

Journal Articles

- [J3] Hearing loss is reflected in conversational pupil responses
Masters, B., Shiell, M.M., Wendt, D., MacDonald, E.N.
Trends in Hearing, 2026
- [J2] Variable delay affects conversational turn-taking behavior in the presence of background noise
Masters, B., MacDonald, E.N.
The Journal of the Acoustical Society of America, 2025
- [J1] Pupil responses during interactive conversation
Masters, B., Aliakbaryhosseinabadi, S., Wendt, D., MacDonald, E.N.
Trends in Hearing, 2025

Conference/Workshop Presentations & Posters

- [C11] Investigating the regulation of speech level during conversations in noise
Masters, B., MacDonald, E.N.
Eriksholm Summer School, Snekkersten, Denmark, 2026
- [C10] The impacts of asymmetric acoustic conditions on speech production and communication dynamics during interactive conversation
Masters, B., Stepanenko P., MacDonald, E.N.
Acoustical Society of America 190, Philadelphia, Pennsylvania, 2026
- [C9] Pupillometry reveals differences in the cognitive demands of interactive conversation versus passive listening
Masters, B., Shiell, M., Wendt, D., MacDonald, E.N.
Acoustical Society of America 189, Honolulu, Hawaii, 2025
- [C8] Assessing the sensitivity of the pupil dilation response to different sources of difficulty in interactive conversation
Masters, B., Stepanenko, P., Shiell, M., Wendt, D., MacDonald, E.N.
Acoustical Society of America 189, Honolulu, Hawaii, 2025

- [C7] Changes in conversational behavior across noise, language proficiency, and task
Stepanenko, P., **Masters, B.**, MacDonald, E.N.
Acoustical Society of America 189, Honolulu, Hawaii, 2025
- [C6] Preliminary investigation of the effect of noise on semantic content of conversations
Stepanenko, P., **Masters, B.**, MacDonald, E.N.
Acoustical Society of America 189, Honolulu, Hawaii, 2025
- [C5] The effects of hearing loss on pupil responses in conversation
Masters, B., Shiell, M., Wendt, D., MacDonald, E.N.
International Symposium on Auditory and Audiological Research, Nyborg, Denmark, 2025
Symposium on Aging Research, Waterloo, Ontario, 2025 (**Research Institute for Aging Award**)
- [C4] Understanding effortful listening in interactive conversation with pupillometry
Masters, B., Shiell, M., Wendt, D., MacDonald, E.N.
Workshop in Communication in Hearing Science, Lyngø, Denmark, 2025
Biomedical Engineering Research Day, Waterloo, Ontario, 2025 (**Best Presentation Award**)
- [C3] Analysis of physiological measures around conversational state changes
Masters, B., Aliakbarhosseinabadi, S., Wendt, D., MacDonald, E.N.
Acoustical Society of America 186/Acoustics Week in Canada, Ottawa, Ontario, 2024
- [C2] The effects of delay on task-based interactive conversation
Masters, B., MacDonald, E.N.
Acoustical Society of America 186/Acoustics Week in Canada, Ottawa, Ontario, 2024
- [C1] Directionality characteristics of the Tympan open-source hearing aid and earpieces
Masters, B., Alexander, J.M.
Acoustical Society of America 181, Seattle, Washington, 2021

Invited Talks

- [I3] Measuring effort in conversation: physiological and linguistic approaches
Technical University of Denmark, 2026
- [I2] Inferring effort in conversation: behavioral and physiological markers
Sonova, 2026
- [I1] Talker sensitivity to turn-taking in conversation
Eriksholm Research Centre, 2024
Technical University of Denmark, 2024

Open Datasets

- [D3] Replication Data for Hearing loss is reflected in conversational pupil responses
Masters, B., Shiell, M.M., Wendt, D., MacDonald, E.N.
Borealis, 2026
- [D2] Replication Data for Variable delay affects conversational turn-taking in the presence of background noise
Masters, B., MacDonald, E.N.
Borealis, 2025
- [D1] Replication Data for Pupil responses during interactive conversation
Masters, B., Aliakbarhosseinabadi, S., Wendt, D., MacDonald, E.N.
Borealis, 2025

Theses

- [T1] Talker Sensitivity to Turn-Taking in Conversation
M.A.Sc. Systems Design Engineering, University of Waterloo, 2024